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PI Name:	Gilbert, Jack Ph.D.		
Project Title:	Quantifying Selection for Pathogenicity and Antibiotic Resistance in Bacteria and Fungi on the ISS – a Microbial Tracking Study		
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PI Email:	gilbertjack@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	630-915-2383
Organization Name:	University of California, San Diego		
PI Address 1:	9500 Gilman Dr		
PI Address 2:	MC 0763		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-5004	**Congressional District:**	52
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Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
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COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
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Task Description:	The built environment (BE) is a perfect model to examine microbial adaptation on the edge of survival. However, the role of the BE in mediating microbial selection remains unquantified. The International Space Station (ISS) provides a unique environment to study microbial dynamics, because it is a sealed system with a standing occupant population of approximately six people and visits spaced by 12-week periods. Notably, this time frame correlates with our previous work, which showed that bacteria acquire antibiotic resistance genes within 6 weeks in a hospital environment. This project will provide insights into the mechanisms of bacterial and fungal ecology, as it pertains to the development of antimicrobial resistance (AMR) and virulence phenotypes in the unique environment of the ISS. By isolating organisms that survive in these environments, quantifying their phenotypes in vitro and in vivo, sequencing their genomes, and leveraging metagenome-based genome reconstruction and ancestral genotype prediction, it should be possible to identify the genetic traits that code those microbial phenotypes (e.g. antibiotic resistance and virulence) that negatively influence human health. In this project, we are testing the hypothesis that built environments select for stress tolerance in human-associated microbes, which result in an increase in (i) genes and pathways associated with antibiotic resistance and pathogenicity, (ii) purifying selection of these traits, (iii) minimum inhibitory antibiotic concentration, and (iv) in vivo pathogenicity of isolated cultures in ISS microbes. We aim to examine this phenomenon in samples collected over a 20-year period that will leverage both the ISS Microbial Tracking database and a new series of collections. Objective 1: To determine the distribution and selection of antimicrobial resistance and virulence genes in bacterial and fungal genomes and metagenomes in the ISS environment. Experiments have been designed to understand genes and pathways that are under purifying selection in the ISS environment. Previous NASA investigations of ISS environmental surfaces have isolated nearly 500 microorganisms, which are archived at Life Science Data Archive (LSDA) for future investigations. Strains were identified using partial 16S rRNA gene sequencing analysis, and hence, taxa were not fully characterized. Furthermore, an additional ~500 strains were collected in Microbial Tracking (MT) projects (MT-1; 202 strains and MT-2; 300 strains) and were phylogenetically identified using the full-length 16S rRNA gene sequence or Internal Transcribed Spacer (ITS) region. As these datasets lack in-depth genomic sequencing data, our ability to perform phylogenomic analysis to predict ancestral genotypes or genes and pathways that are under purifying selection in the ISS environment has been limited. This study leverages these two existing NASA culture collections, whereby strains are selected for analysis based on phylogeny, pathogenicity, etc. Selected isolates are being genome sequenced and the associated shotgun metagenomic sequencing is being mined. Tasks include: Task 1. Sequence whole genomes of ISS microorganisms Task 2. Generate bioinformatic antibiotic resistance profiles of ISS microorganisms Task 3. Elucidate genes associated with host-invasion potential of ISS microorganisms Task 4. Assemble genomes from metagenomic data Task 5. Predict ancestral genotypes of key bacterial species Objective 2: To quantify virulence and antimicrobial resistance phenotypes in archived ISS microbial isolates. While there is substantial evidence to suggest increased antibiotic resistance and virulence in space-flown isolates when compared to ground-based organisms, the extent to which the presence of resistance or virulence genes associates with the phenotype in vivo is unclear. We will determine how microbial genotypes, identified in Aim 1, impact AMR and virulence. Up to 200 microbial strains from LSDA, MT-1 and MT-2 collections will be cultured and subjected to detailed antibiotic resistance assays. We will also select 50 strains in which to validate and quantify virulence traits in vivo in nematode mortality models, which our preliminary data suggest are highly sensitive in detecting strain-specific virulence. The results of these assays will be correlated with genetic evidence of antibiotic resistance genes and virulence pathways. Whereas genes predicting resistance or virulence that fail to provide the predicted phenotype will be considered a “false positive”, genotypes that do not predict resistance or virulence will be considered a “false negative”. False negative resistance and virulence genes present potential new therapeutic targets. Tasks include: Task 6. Perform in vitro antibiotic resistance assays Task 7. Perform in vivo virulence assays using C. elegans Objective 3: To utilize live-dead cell characterization methods to determine the distribution and temporal dynamics of living microorganisms, as well as antimicrobial resistance and virulence phenotype in the ISS environment. Less than 30% of intact, potentially viable microorganisms isolated from previous MT flights could be cultured, which has made comprehensive characterization of bacterial and fungal species, and potentially critical genomic changes to AMR and virulence genes, impossible to discern. We have previously developed and deployed propidium monoazide (PMA) treatment to allow molecular characterization of AMR and virulence genes in living cells, which we have shown correlates with phenotypic susceptibility and WGS data of cultured isolates. By applying metagenomics to PMA-treated samples, we plan to capture genotype assemblies from a broader phylogenetic diversity than can be retrieved from culturing alone. In objective three, we have planned a new flight series to use PMA-based molecular analysis, in combination with traditional methods. Phylogenetic analysis, metagenomic sequencing, and phenotypic assays will be performed as in Objectives 1 and 2, and changes in AMR and virulence will be compared to MT-1 and MT-2 missions. MT-1, MT-2, and new data collected will be compiled for analysis to generate a model of AMR and virulence dynamics in the ISS. Tasks include: Task 8. ISS environmental sampling Task 9. Estimate and identify culture-based microbial burden and diversity Task 10. Perform metagenomic sequencing and qPCR of PMA-treated DNA to examine the viable phylogenetic and functional diversity Task 11. Correlating genomic and phenotypic profiles with ISS surface and atmospheric data Task 12. Evaluate virulence phenotypes and antimicrobial resistance in new isolates
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Built environment-associated microbes are predominantly sourced from building occupants, which is restricted to humans on the ISS. Due to this, we can elucidate the contribution and consequences of environment-driven genomic changes that will be critical for the future design and operation of both space and terrestrial built environments in ways that promote ‘healthy’ microbial ecosystems. Further, long-term identification of microbial taxa that pose a threat to crew health can be targeted to develop methods for removal or antimicrobial treatment strategies. This can protect not only crew members from ISS-specific pathogens, but also to tackle earth-bound antibiotic resistance, which has become a world-wide health crisis.

Task Progress:	We whole genome sequenced (WGS) 320 bacterial and 164 fungal isolates from repositories of previously collected microbes, including those from the MT-1 (224 bacterial and 70 fungal) and MT-2 (96 bacterial and 94 fungal) missions (Task 1). Previously, these samples had only been phylogenetically characterized through sequencing of the 16S rRNA gene or Internal Transcribed Spacer (ITS) region. Through WGS, we have characterized genes of interest within individual organisms from different International Space Station (ISS) locations (n = 8) over time. Genes of interest include those associated with pathogenicity, such as those that confer antibiotic resistance and virulence potential (Tasks 2 and 3). While there is substantial evidence to suggest increased antibiotic resistance and virulence in space-flown isolates when compared to ground-based organisms, the extent to which the presence of these genes associates with the phenotype is unclear. Therefore, we conducted antibiotic susceptibility and virulence testing on 48 and 50 microbial isolates, respectively, with special focus on the identification of false positive and false negative results (Tasks 6 and 7). We used the Kirby-Bauer Disk Diffusion Susceptibility Test Protocol for antibiotic resistance assays and a C. elegans model for virulence testing. In addition, we obtained metagenomes from the MT-1 (n = 42) and MT-2 (n = 40) missions for characterization of taxonomy, antibiotic resistance, and virulence genes (Task 4). Metagenomic samples were collected at the same locations/times as the bacterial and fungal isolates; and therefore, have allowed us to compare isolate data to the genetic potential of the whole community. We then collected samples as part of a new flight series (MT-3). Samples were collected during three separate flights (SpX 22 – 24) between June 2021 and January 2022 (Task 8) from the same eight surfaces sampled during the MT-1 and MT-2 missions. However, rather than a single collection timepoint, samples were collected for five consecutive days at the end of each flight (n = 65 with in-flight and ground controls). As with MT-1 and MT-2 samples, we used propidium monoazide (PMA)-based metagenomics to characterize phylogeny, antibiotic resistance, and virulence gene proportions (Task 10), which will be compared to culture-based genomics, and validated by antibiotic resistance and virulence assays (Task 12). For antibiotic resistance and virulence assays, 50 bacterial isolates were selected, based on spatial and taxonomic variation. Further, we estimated microbial burden with quantitative polymerase chain reaction (qPCR) analysis (Task 9). These data have allowed us to better understand the environmental factors that influence microbial survival and distribution on spacecraft (Task 11). We found spatial and temporal differences in the relative abundance of organisms, as well as trends in the distribution of genes associated with antibiotic resistance and virulence.
Bibliography Type:	Description: (Last Updated: 11/18/2025)
Articles in Other Journals or Periodicals	Simpson AC, Sengupta P, Zhang F, Hameed A, Parker CW, Singh NK, Miliotis G, Rekha PD, Raman K, Mason CE, Venkateswaran K. "Phylogenetic affiliations and genomic characterization of novel bacterial species and their abundance in the International Space Station." Res Sq [Preprint]. 2023 Jul 6:rs.3.rs-3126314. https://doi.org/10.21203/rs.3.rs-3126314/v1 ; PMID: 37461605; PMCID: PMC10350232 , Jan-2024
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